

Section A — What is already known: GaAs and the incorporation of Bi

$\text{GaAs}_{1-x}\text{Bi}_x$ is the dilute bismide formed by alloying bismuth into the binary III–V semiconductor GaAs. In the zincblende structure GaAs comprises two interpenetrating face-centred-cubic sublattices: a gallium (group-III, cation) sublattice and an arsenic (group-V, anion) sublattice. Bismuth is itself a group-V element, so its incorporation is Isovalent, a Bismuth atom substitutes for an Arsenic atom on the anion sublattice rather than for Ga, leaving the cation sublattice unchanged. Tixier et al. (2003) infer this indirectly rather than measuring it; they observe that Bi incorporation requires the arsenic-to-gallium flux ratio to be held close to stoichiometry, and interpret the resulting competition between the two group-V species for the same sites as evidence that Bi occupies As sites (Tixier et al., 2003, p. 2245). The source frames this as an assumption supported by growth data, not a direct structural determination, a distinction worth preserving, because the alloy's electronic structure and strain behaviour — developed later in this dissertation, both rest on Bi sitting on the As sublattice.

The defining property of the substituting atom is its size. Bismuth's large atomic size governs the growth chemistry: under MBE conditions Bi surface-segregates and is largely rejected from the film, so significant incorporation proceeds only at a much reduced substrate temperature (380 °C) and with a near-stoichiometric As:Ga flux ratio (Tixier et al., 2003, p. 2245). The oversized atom constrains the entire growth window rather than acting as a passive dopant.

The introduction of an oversized group-V atom into the As sublattice is solely responsible for both the advantageous properties and inherent limitations of the alloy. Electronically, substituting As with Bi reduces the band gap significantly: GaBi is predicted to be a semimetal (potentially find another paper to confirm this as the use of the word 'predicted' probably isn't suited), so even dilute Bi produces a large redshift, measured by Tixier et al. (2003, p. 2246) at approximately 84 meV per percent Bi. This exceeds the band-gap shifts produced by indium or antimony substitution at the same alloy fraction (≈ 16 meV/% In in InGaAs and ≈ 21 meV/% Sb in GaAsSb) and is of the same order as dilute-nitrogen alloying (≈ 125 meV/% N in GaNAs) (Tixier et al., 2003, p. 2246). Structurally, the same oversized atom expands the lattice in proportion to Bi content, following Vegard's law, with the GaBi end-point extrapolating to 6.33 ± 0.06 Å and a corresponding $\approx 12\%$ mismatch against GaAs a misfit strain that grows with x and, in thicker high-Bi layers, may begin to relax plastically through structural disorder and misfit dislocations (Tixier et al., 2003, pp. 2245–2247). The benefit— band-gap reduction for infrared optoelectronics and the cost— epitaxial strain are thus two consequences of a single cause.

BEAT(1)- Mechanism

Bismuth substitutes for arsenic on the group-V sublattice as an isovalent impurity. Because Bi has a significantly lower electronegativity and ionisation energy than As, the localised defect states it introduces lie near the valence-band edge of GaAs rather than deep within the gap

(Alberi et al., 2007). These are T_2 -symmetric states: four primary p-like levels roughly 0.4 eV below the valence-band maximum (VBM), and two spin-orbit split-off levels about 2.6 eV below it.

The band-gap reduction follows from how these states couple to the host. In the valence-band anti-crossing (VBAC) model, the localised T_2 states of the minority Bi atoms hybridise with the extended states of the GaAs valence band, restructuring the heavy-hole (HH), light-hole (LH) and spin-orbit split-off (SO) bands into E_+ and E_- subbands (Alberi et al., 2007). The E_- subbands keep a considerable amount of localised character inherited from the Bi defect states, while the E_+ subbands hold much of the extended character of the host. The coupling strength is set by an adjustable parameter C_{Bi} , which depends on the ionisation-energy difference between As and Bi and is fitted to the data; a value of $C_{Bi} = 1.6\text{eV}$ reproduces the measured transition energies.

The consequence that matters mechanistically is a nonlinear upward shift of the HH/LH E_+ band edges as x increases, and Alberi et al. (2007) identify this movement of the valence-band edge as the primary cause of band-gap bowing in $\text{GaBi}_x\text{As}_{1-x}$. The observable result is an initial band-gap decrease of approximately 90 meV per $x = 0.01$. The same anticrossing also renders the dispersion distinctly nonparabolic, and the broadening of the HH E_+ band is expected to raise the hole effective mass and strongly influence hole transport in the alloy.

Beat(1)- Payoff one

The motivation for a temperature-insensitive band gap is a device limitation. Oe (2002) says that the lasing wavelength of GaInAsP semiconductor lasers in wavelength-division-multiplexing (WDM) systems drifts with ambient temperature, because both the band gap and the refractive index depend on temperature. Holding the wavelength fixed therefore requires Peltier elements that stabilise the laser temperature through current control — an overhead that an active layer with a temperature-insensitive band gap would remove, making WDM systems, in Oe's framing, more practical and cheaper.

$\text{GaAs}_{1-x}\text{Bi}_x$ supplies that behaviour. From PL measurements over 10–300 K, Oe (2002) finds the peak energy shifting to longer wavelengths as Bi content rises, with a temperature dependence far weaker than the band-gap variation of GaAs: for a $\text{GaAs}_{0.974}\text{Bi}_{0.026}$ layer the PL peak energy moves by only 0.1 meV/K, less than one-third of the GaAs value.

Photoreflectance measurements confirm that these PL peak energies coincide with the true band-gap energies, so the weak shift reflects the gap itself rather than another transition.

Oe (2002) interprets this through a gap-coefficient argument. Following Oe and Asai, the proposal is that combining a conventional semiconductor, which has a negative band-gap temperature coefficient, with a positive-coefficient component lets the two opposing dependences offset. That positive coefficient is associated with a band-inversion, semimetallic character: in $\text{Hg}_{1-x}\text{Cd}_x\text{Te}$ the coefficient is positive below $x = 0.6$, and the gap of $\text{Hg}_{0.4}\text{Cd}_{0.6}\text{Te}$ does not change with temperature. Bi is put forward as the semimetallic partner,

since Bi and some of its compounds are semimetallic like HgTe. Oe frames this as a hypothesis that the GaAsBi results support, not as an established mechanism.

Beat(1)- Payoff two

Spin–orbit engineering and Auger suppression

A second payoff follows from the spin–orbit-splitting energy ΔSO , which grows as the band gap shrinks. Usman *et al.* (2011) trace this to the same valence-band anticrossing that drives the gap reduction: the Bi-related resonant level interacting with the heavy- and light-hole edge pushes the GaAs valence-band edge (VBE) upward, and since $\Delta\text{SO} = E_{\text{v1}} - E_{\text{SO}}$ is set by that edge relative to the split-off band, the upward VBE shift inflates ΔSO directly (Usman *et al.*, 2011). In their tight-binding calculations this VBE upshift is steep at low Bi content (~ 53 meV per % Bi, easing to ~ 30 meV per % Bi at higher x) and dominates the ΔSO bowing, whereas the split-off edge itself falls only gently and linearly (~ 5.5 meV per % Bi) through conventional alloying (Usman *et al.*, 2011). The gap and ΔSO therefore move in opposite directions and cross at approximately 10.5% Bi — a crossover their model reproduces in agreement with photoreflectance measurements (Usman *et al.*, 2011). Beyond it the alloy occupies an $E_{\text{g}} < \Delta\text{SO}$ regime, which is where the device relevance lies: in this regime the CHSH Auger channel that Usman *et al.* (2011) describe — a conduction electron and heavy hole recombining to excite a hole from the heavy-hole band into the split-off band — is expected to be suppressed, removing a non-radiative loss that otherwise dominates the threshold current of telecom-wavelength III–V lasers. Usman *et al.* (2011) establish the crossover computationally; the suppression itself they report as a suggested, and therefore composition-contingent, benefit realised only above $\sim 10.5\%$ Bi.

Beat 1 — Applications: low-noise avalanche photodiodes

Avalanche photodiodes (APDs) amplify weak optical signals by impact ionisation, but the process is stochastic, so each multiplication event carries excess noise that grows with gain and sets a ceiling on usable sensitivity (Liu et al., 2021). Where that ceiling sits is fixed by the ratio of the electron and hole ionisation coefficients, α/β : the noise is lowest when one carrier overwhelmingly drives the avalanche, equivalently when the excess-noise factor $k = \beta/\alpha$ in McIntyre's relation is small (Liu et al., 2021).

Conventional III–V alloying cannot deliver a large α/β , because the two coefficients move together. Liu et al. (2021) note that the ratio barely shifts with composition in InGaAsP, AlGaAs or InGaAs/GaAs superlattices, and that adding aluminium to GaAs raises the electron and hole ionisation fields by similar amounts. The reason is structural: the carriers that actually ionise do not sit at the band edges but in higher-lying bands at much greater energy, so engineering the band edges alone leaves the ionisation physics nearly untouched (Liu et al., 2021).

Bi moves one coefficient without the other, and it does so through the spin–orbit payoff established above. Holes in GaAs reach their ionisation threshold mainly by scattering out of

the heavy- and light-hole bands into the split-off band; raising ΔSO pushes that band out of reach, suppressing β , while α , governed by the conduction-band side, stays close to its GaAs value (Liu et al., 2021). This route does not require ΔSO to exceed E_g . The $\Delta SO > E_g$ condition that suppresses the Auger channel in lasers needs more than 10% Bi, beyond the composition where GaAsBi remains growable as high-quality material; the APD effect needs only that ΔSO increase, so it is already strong at the few-percent compositions that can be grown well (Liu et al., 2021).

The Sheffield group demonstrated this across thirteen p-i-n and n-i-p diodes with up to 5.1% Bi (Liu et al., 2021). As Bi is added, α falls by only about a third relative to GaAs while β drops by orders of magnitude at low field; the net result is a 2- to 100-fold rise in α/β (Liu et al., 2021). The hole-ionisation threshold field climbs from 217 kV/cm in GaAs to 333 kV/cm at 5.1% Bi and tracks the measured ΔSO increase of roughly 46.7 meV per %Bi between 1 and 11% Bi, whereas the electron threshold stays near 210 kV/cm (Liu et al., 2021). The excess noise follows the ratio: the effective k drops from 0.48 in GaAs to 0.2 at 4% Bi, and by 3.5% Bi the excess noise approaches that of silicon, while the absorption edge extends past 1.1 μm into the near-infrared (Liu et al., 2021). Liu et al. (2021) emphasise that a one-sided change of this magnitude in β alone has not been observed in any other dilute alloy and appears specific to Bi.

This is the evidence Beat 1 needs. The benefit here is not a calculated band structure but a fabricated, measured device in which a few percent Bi converts a predicted ΔSO shift into a quantified noise reduction, and it pays out at the low-Bi end where the alloy stays growable — the same front-loading of benefit at low x on which the bowing argument rests. Liu et al. (2021) bound the claim honestly: antimony, another large group-V atom, can raise ΔSO comparably, and equivalent Sb structures still record marginally lower excess noise. GaAsBi is therefore a leading demonstration of valence-band engineering for low-noise multiplication rather than a proven optimum.

Beat 2 — cost (lattice strain)

The lattice expansion set out above cannot relax freely once GaAsBi is grown epitaxially on GaAs. The relaxed alloy constant is larger than the substrate's and grows further with x , so a coherent epilayer must compress its in-plane lattice to register with the GaAs template, placing it under biaxial compression. The cell relieves part of this by stretching along the growth axis — the tetragonal distortion that Tixier et al. (2003, p. 2246) correct for explicitly, recovering the free-standing lattice parameter from the measured growth-direction value using the elastic constants of GaAs. Their asymmetric reciprocal-space maps confirm that the 0.4%, 1.3% and 3.1% layers grow pseudomorphically (Tixier et al., 2003, p. 2246): the misfit is stored as elastic strain rather than released, so the film carries the full mismatch.

That stored strain is negligible when Bi is dilute but climbs steeply with composition. The 3.1% layer is mismatched to GaAs by 0.37% (Tixier et al., 2003, p. 2246); a 20% layer carries 2.4% (Lewis et al., 2012, p. 082112-2) — both consistent with the linear Vegard scaling the two groups assume, at close to 0.12% of misfit for each percent of Bi (**Assumption made off statistics from paper**). Because elastic strain energy scales with the

square of the misfit, the stored energy, and with it the thermodynamic driving force for relaxation, rises faster than x itself.

Beyond a critical thickness the elastic energy stored in the strained layer outweighs the energy needed to introduce misfit dislocations, and relaxation becomes the lower-energy state. The two studies bracket this threshold from opposite sides. Tixier et al. (2003, p. 2246) note that InGaAs of comparable mismatch relaxes above a critical thickness near 60 nm, far below the 270 nm of their 3.1% layer; and although their maps still classify that layer as pseudomorphic, its diffraction tells a more ambiguous story — the epilayer reflection has lost its thickness fringes and sits on a broad diffuse-scattering peak, which the authors read as structural disorder, Bi concentration inhomogeneity, and possibly the onset of plastic deformation through misfit dislocations (Tixier et al., 2003, p. 2246). Lewis et al. (2012, p. 082112-2) sit at the other extreme: a 20 nm layer at 20% Bi, whose equilibrium critical thickness on the Matthews–Blakeslee criterion is under 5 nm, is found by reciprocal-space mapping to be fully strained. A coherent film more than four times thicker than its equilibrium limit is metastable — the dislocations that would relax it have not nucleated and propagated.

This contrast is resolved by kinetics, not thermodynamics. Misfit dislocation nucleation and glide are thermally activated, and the two groups grow in very different thermal regimes: Tixier et al. (2003, p. 2245) reach 3.1% Bi at 380 °C, whereas Lewis et al. (2012) attain 22% only by lowering the substrate to 200 °C. The low temperature that high Bi incorporation requires happens also to freeze out the relaxation pathway, letting thin layers hold strain they would shed if grown warm. The same requirement, though, is what closes the growth window as x rises. Incorporated Bi is weakly bound — Lewis et al. (2012, p. 082112-2) measure an activation energy near 0.25 eV for the temperature dependence of incorporation and observe Bi content climbing sharply as the substrate cools — so it segregates to the surface and is thermally re-ejected unless the substrate is kept cold. Maximising Bi additionally forces the flux into the As-poor, Ga-rich corner, where Ga and Bi droplets form on the growth front (Lewis et al., 2012, p. 082112-2). Raising x therefore tightens several constraints at once: a smaller equilibrium critical thickness, a colder and more flux-sensitive growth condition, and an increasing tendency toward droplets and compositional inhomogeneity.

Strain leaves its mark even where coherence survives. The photoluminescence linewidth in Tixier et al. (2003, p. 2246), used here only as a measure of structural quality, broadens steadily with Bi content — from about 50 meV at 0.4% to 106 meV at 3.1%, against under 40 meV for comparable InGaAs — and the authors attribute it jointly to the strain energy of the high-Bi layers and to the disorder imposed by the low-temperature, low-As₂ growth that Bi incorporation forces (Tixier et al., 2003, pp. 2246–2247). The two contributions share a root: the oversized atom that loads the lattice with misfit also dictates the cold, narrow conditions under which crystalline order degrades. The response Tixier et al. (2003, p. 2247) reach for is geometric restriction — confining GaAsBi to quantum-well-scale thicknesses to stay below the relaxation limit — and the coherent high-Bi layers of Lewis et al. (2012, p. 082112-2), only tens of nanometres thick, are that strategy in practice.

Beat 3 — Synthesis: the trade-off verdict

Taken together, the preceding beats describe one substitution read at two ends of the composition axis. Richards et al. (2022) state both ends: strong band engineering is available from a few percent bismuth, while progress toward the devices that need more has been held back by the material-quality problems that follow from the low growth temperatures large Bi fractions require (Richards et al., 2022, p. 2100330-1). These usually read as two separate observations about the alloy. They are better taken as a single trade-off, and its structure is what makes GaAsBi worth the difficulty of growing it: the benefit is concentrated at low x and the cost at high x , so a dilute layer banks most of the band-structure change before the strain and growth penalties become limiting.

That the trade is favourable, rather than merely real, follows from the shapes of the two curves against x . The band-gap reduction is front-loaded: the shift per unit bismuth is steepest as the first atoms enter and slackens as x climbs. Beats 0 and 1 establish exactly this, from the large low- x redshift and a valence-band edge that moves fastest at low Bi and eases at higher content; most of the accessible band-structure change is therefore secured at low composition. The cost accelerates the other way, as Beat 2 sets out: strain energy scales with the square of the misfit and the misfit grows with x , so the drive toward relaxation climbs as x^2 or steeper. It steepens further if the positive deviation from Vegard's law that Madouri et al. (2008, p. 820) calculate holds at high composition, where the lattice expands beyond the linear scaling Beat 2 assumes for the dilute regime. The benefit slope is therefore largest exactly where the cost is smallest, and the ratio between them falls as bismuth is added.

The verdict holds only inside the dilute window, and that limit is its substance rather than a caveat to it. The devices that sit there take the front-loaded benefit directly: the near-infrared detectors and low-noise avalanche diodes of Beat 1 work at a few percent Bi, below the composition where crystal quality degrades. The ones Richards et al. (2022) single out as held back sit outside it. Suppressing the CHSH Auger channel in a laser needs $\Delta SO > E_g$, and so upward of 10% Bi (Beat 1), which is squarely in the region where the sub-5 nm critical thickness and droplet-prone growth window of Beat 2 act together. For that class the benefit is not front-loaded at all; the full cost must be paid to reach the composition where it becomes useful. GaAsBi is therefore a favourable alloy for the devices that can exploit dilute bismuth and a marginal one for those that cannot, and it is the composition dependence of the penalty, not of the benefit, that decides which side a given device falls on.

References

Tixier, S., Adamcyk, M., Tiedje, T., Francoeur, S., Mascarenhas, A., Wei, P. and Schiettekatte, F. (2003) 'Molecular beam epitaxy growth of GaAs_{1-x}Bi_x', *Applied Physics Letters*, 82(14), pp. 2245–2247.
available at: doi: 10.1063/1.1565499.

Alberi, K., Dubon, O.D., Walukiewicz, W., Yu, K.M., Bertulis, K. and Krotkus, A. (2007) 'Valence band anticrossing in $\text{GaBi}_x\text{As}_{1-x}$ ', *Applied Physics Letters*, 91(5), p. 051909. Available at: <https://doi.org/10.1063/1.2768312>.

Oe, K., 2002. Characteristics of Semiconductor Alloy $\text{GaAs}_{1-x}\text{Bi}_x$. *Japanese Journal of Applied Physics*, 41(5A), pp.2801-2806. available at: <https://doi.org/10.1143/JJAP.41.2801>

Usman, M., Broderick, C.A., Lindsay, A. and O'Reilly, E.P. (2011) 'Tight-binding analysis of the electronic structure of dilute bismide alloys of GaP and GaAs', *Physical Review B*, 84(24), 245202. doi: 10.1103/PhysRevB.84.245202.

Link: <https://doi.org/10.1103/PhysRevB.84.245202>

Liu, Y., Yi, X., Bailey, N.J., Zhou, Z., Rockett, T.B.O., Lim, L.W., Tan, C.H., Richards, R.D. and David, J.P.R. (2021) 'Valence band engineering of GaAsBi for low noise avalanche photodiodes', *Nature Communications*, 12, 4784.

Link: <https://doi.org/10.1038/s41467-021-24966-0>.

Lewis, R.B., Masnadi-Shirazi, M. and Tiedje, T. (2012) 'Growth of high Bi concentration $\text{GaAs}_{1-x}\text{Bi}_x$ by molecular beam epitaxy', *Applied Physics Letters*, 101(8), 082112. doi: 10.1063/1.4748172.

Richards, R.D., Bailey, N.J., Liu, Y., Rockett, T.B.O. and Mohmad, A.R. (2022) 'GaAsBi: from molecular beam epitaxy growth to devices', *Physica Status Solidi B*, 259(2), 2100330. doi: 10.1002/pssb.202100330.

Madouri, D., Boukra, A., Zaoui, A. and Ferhat, M. (2008) 'Bismuth alloying in GaAs: a first-principles study', *Computational Materials Science*, 43, pp. 818–822. doi: 10.1016/j.commatsci.2008.01.059.